MIL-P-46416A(MU)
12 APRIL 1964
SUPERSEDING
MIL-P-46416(ORD)
15 NOVEMBER 1960

MILITARY SPECIFICATION

PROJECTILE, 25 MILLIMETER, M379 FOR TRAINER MORTAR M32, ASSEMBLY

1. SCOPE

1.1 This specification covers the assembly and packing for one type of projectile designate as Projectile. 25MM. M379 for Trainer Mortar M32.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-320 — Chromium Plating (electrode - deposited).

QQ-P-416 — Plating, Cadmium (electrode - deposited).

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-P-10025 — Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for.

MIL-I-45607 — Inspection Equipment Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8845009 — Projectile, 25MM, M379, for Trainer Mortar M32, Assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:54Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

ORDNANCE CORPS

IEL–8845009 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (dwg.) 8845009 and with all requirements in applicable specifications.

3.3 Profile and alignment. The assembly shall freely enter the profile and alignment gage when tested as specified in 4.4.1.

3.4 Embrittlement of projectile spring. The projectile spring (see dwg. 8845013) shall not break when extended to the minimum (min.) point of permanent distortion when tested as specified in 4.4.2.

3.5 Workmanship. All parts shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. In addition, the plastic parts shall be free of porosity warpage, checks, cracks, chipped edges, blisters, excessive flash and other defects affecting their serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define terms used herein. The provisions of MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:54Z
Check the source to verify that this is the current version before use.

The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawing and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the contracting officer (see 6.5) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of 15 consecutively produced complete assemblies and 25 complete sets of parts (i.e. 25 each of every component) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 *Inspections to be performed.* Each assembly shall be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly or subassembly fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly or subassembly in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of projectile assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:54Z
Check the source to verify that this is the current version before use.

specification revision. Drawing, specification, and process changes not affecting safety, performance or fit as determined by the Government shall not necessitate changing the lot number. Product shall be submitted for inspection in accordance with MIL–STD–105 (or MIL–STD–1235, when applicable).

4.3.2 *Examination.* Inspection for critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Firing pin (see dwg. 8845014 covering a detail of dwg. 8845009).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical: None defined.			
Major:	AQL 1.00 percent		
101.	Concentricity of recess diameter at small end with large outside diameter	Gage	01001
102.	Concentricity of small outside diameter with large outside diameter	Gage	01002
103.	Small outside diameter	Gage	01003
104.	Diameter of recess at small end	Gage	01004
105.	Depth of recess at small end	Gage	01005
106.	Total length	Gage	01006
107.	Distance from large end to shoulder	Gage	01007
108.	Large outside diameter	Gage	01008
Minor:	AQL 1.00 percent		
201.	Protective finish with bare spot	Visual	01009
202.	Evidence of poor workmanship (see 3.5)	Visual	01010

4.3.2.2 Nose (see dwg. 8845010 covering a detail of dwg. 8845009).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 1.00 percent		
101.	Concentricity of center hole diameter with pitch diameter of thread	Gage	02001
102.	Perpendicularity of flange face with pitch diameter of thread	Gage	02002
103.	Diameter of center hole	Gage	02003
104.	Pitch diameter of thread, min.	Gage	02004
105.	Major diameter of thread, min.	Gage	02005
106.	Transverse holes missing	Visual	02006
Minor:	AQL 1.50 percent		
201.	Outside diameter of flange, max.	Gage	02007
202.	Any hole obstructed	Visual	02008
203.	Protective finish with bare spot	Visual	02009
204.	Evidence of poor workmanship (see 3.5)	Visual	02010

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:54Z
Check the source to verify that this is the current version before use.

4.3.2.3 Retaining ring (see dwg. 8845012 covering a detail of dwg. 8845009).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	**AQL 0.65 percent**		
101.	Diameter of hole	Gage	03001
102.	Concentricity of outside diameter with hole	Gage	03002
103.	Outside diameter	Gage	03003
Minor:	**AQL 1.50 percent**		
201.	Recess missing	Visual	03004
202.	Protective finish with bare spot	Visual	03005
203.	Evidence of poor workmanship (see 3.5)	Visual	03006

4.3.2.4 Projectile spring (see dwg. 8845013 covering a detail of dwg. 8845009).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	**AQL 0.40 percent**		
101.	Load at compressed length	Gage	04001
102.	Load at assembled length	Gage	04002
103.	Inside diameter	Gage	04003
104.	Outside diameter, max.	Gage	04004
Minor:	**AQL 1.00 percent**		
201.	Protective finish with bare spot	Visual	04005
202.	Evidence of poor workmanship (see 3.5)	Visual	04006

4.3.2.5 Insert ring, prior to molding into body assembly (see dwg. 8845011 covering a detail of dwg. 8845009).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	**AQL 1.00 percent**		
101.	Concentricity of hole diameter with pitch diameter of thread including perpendicularity of threaded end	Gage	05001
102.	Concentricity of retaining ring cavity with pitch diameter of thread including perpendicularity of threaded end	Gage	05002
103.	Concentricity of counterbore with pitch diameter including perpendicularity of the threaded end	Gage	05003
104.	Pitch diameter of thread, max.	Gage	05004
105.	Minor diameter of thread, max.	Gage	05005
106.	Any operation missing	Visual	05006
107.	Crack	Visual	05007
Minor:	**AQL 1.00 percent**		
201.	Protective finish with bare spot	Visual	05008
202.	Evidence of poor workmanship (see 3.5)	Visual	05009

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:54Z
Check the source to verify that this is the current version before use.

4.3.2.6 Body assembly (see dwg. 8845015 covering a detail of dwg. 8845009).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	**AQL 0.65 percent**		
101.	**Diameter of bourrelets**	Gage	06001
102.	**Length**	Gage	06002
103.	**Molding incomplete**	Visual	06003
Minor:	**AQL 1.00 percent**		
201.	**Insert above or excessively below flush**	Gage	06004
201.	**Evidence of poor workmanship (see 3.5)**	Visual	06005

4.3.2.7 Assembly (see dwg. 8845009).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	**AQL 0.65 percent**		
101.	**Total length, max.**	Gage	07001
102.	**Component missing, incorrectly assembled or obviously damaged**	Visual	07002
103.	**Cracks in any area**	Visual	07003
104.	**Firing pin not free to move within retaining ring (see 6.4)**	Visual-Manual	07004
Minor:	**AQL 1.00 percent**		
201.	**Molded character missing or unidentifiable**	Visual	07005
202.	**Evidence of poor workmanship (see 3.5)**	Visual	07006

4.3.3 *Testing.*

4.3.3.1 *Profile and alignment (see 3.3) — Major defect — Code No. 07007.* The assemblies shall be gaged 100 percent for profile and alignment using equipment in accordance with 4.3.4. The test shall be performed as specified in 4.4.1.

4.3.3.2 *Embrittlement of projectile spring (see 3.4) — Major defect — Code No. 04007.* The sampling plan for this test shall be in accordance with MIL–STD–105, Code Letter G and an AQL of 0.40 percent using equipment in accordance with 4.3.4. The test shall be performed as specified in 4.4.2.

4.3.3.3 *Plating thickness and adhesion tests of nose (see dwg. 8845010) — Minor defects — Code Nos. 02011 and 02012.* The sampling plan, test procedure and acceptance criteria for these tests shall be in accordance with Specification QQ–C–320.

4.3.3.4 *Plating thickness, adhesion and salt spray tests of insert, projectile spring, retaining ring and firing pin (see applicable drawings) — Minor defects — Code No. 05010, 04008, 03007 and 01011.* The sampling plan, test procedure and acceptance criteria for these tests shall be in accordance with Specification QQ–P–416.

4.3.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–8845009 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:54Z
Check the source to verify that this is the current version before use.

and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Profile and alignment.* Each projectile assembly shall freely enter the profile and alignment gage. Any assembly which fails to meet the requirement of 3.3 shall be classed defective and removed from the lot.

4.4.2 *Embrittlement of projectile spring.* The projectile spring shall be pulled to the minimum point of permanent distortion. Observation shall be made for compliance with the requirement specified in 3.4. Springs used in this test shall not be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 *Level C.* The projectile assemblies shall be packed and marked in accordance with MIL–P–10025.

5.2 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showings in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:54Z
Check the source to verify that this is the current version before use.

accordance with MIL–D–45608

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black line on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:54Z
Check the source to verify that this is the current version before use.

final design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Firing pin movement. Inspection for this condition will be accomplished by tipping the assembly from base end down to nose end down while observing the movement of the firing pin. If there is no perceptible movement of the firing pin within the retaining ring, it will be assumed that the firing pin is bound because of dimensional deviation, that is, the shank diameter tapers beyond the maximum permitted, the diameter of the hole in the retaining ring is undersize or that the diameter of the firing pin shank and the larger diameter of the firing pin are not sufficiently concentric with each other. If the shank of the firing pin travels through the retaining ring for a distance of approximately .2 inch, it will be assumed that the projectile spring is missing.

6.5 Submission of initial production samples. Instructions as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–NI, Dover, N.J.

Custodian:
Army—MU

Preparing activity:
Army—MU

Project No. 1305–A340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:54Z
Check the source to verify that this is the current version before use.